

Conservative Open-Vocabulary Segmentation: Grounded Proposals, Alignment Gating, and CLIP-Guided Iterative Refinement

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Abstract—Open-vocabulary semantic segmentation requires identifying and delineating arbitrary object regions described by free-form text prompts, without restriction to a fixed training vocabulary. We present a five-stage grounded segmentation pipeline that integrates GroundingDINO for text-conditioned bounding-box proposals, SAM2 for high-quality binary mask generation, CLIP (ViT-B/32) for zero-shot semantic scoring, a *Conservative Alignment MLP* that selectively re-ranks mask candidates only under multi-signal uncertainty, a *Gradient-Boosted Tree Gate+Rank Policy* for one-step candidate switching, and an *Iterative CLIP-Guided Refiner* that re-prompts SAM2 with distance-transform foreground points and applies Dense CRF boundary correction under a strict semantic accept-or-revert rule.

Our central finding is that iterative refinement *without* intelligent gating is actively harmful—dropping mIoU by up to 5.6 pp on PASCAL Context-59 and 3.7 pp on PASCAL VOC—while gated refinement preserves or improves quality. An oracle experiment reveals that the dominant bottleneck is candidate *selection*, not mask generation: the proposal pool already contains 0.580, 0.483, 0.720, 0.546, and 0.529 mIoU solutions on COCO-Object, COCO-Stuff, PASCAL VOC, PASCAL Context-59, and ADE20k respectively. Our tree policy recovers 13–47% of the oracle gap over the CLIP baseline. Evaluated across five benchmarks in a zero-shot setting, the full pipeline achieves 37.6 mIoU on COCO-Stuff and 45.4 mIoU on ADE20k, surpassing prior state-of-the-art training-free composable methods by 9.5 pp and 24.3 pp respectively, while remaining competitive (66.7 mIoU) on PASCAL VOC.

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Index Terms—open-vocabulary segmentation, GroundingDINO, SAM2, CLIP, conservative alignment, gradient-boosted policy, iterative refinement, CRF, zero-shot segmentation

I. INTRODUCTION

Semantic segmentation assigns a class label to every pixel in an image. Classical closed-set models train a fixed decoder for a predefined taxonomy and fail gracefully outside it [16]–[18]. The emergence of large-scale vision–language foundation models opens a different pathway: *open-vocabulary segmentation* (OVS), where pixel predictions are driven by arbitrary natural-language descriptions without category-specific retraining.

Two complementary capabilities are now available off-the-shelf. CLIP [1] provides a joint embedding space for images and text that enables zero-shot semantic scoring of any image region against any textual concept. GroundingDINO [4] provides open-set text-conditioned bounding-box proposals. SAM2 [3] converts box prompts into high-quality binary masks. Naively chaining these three tools—using GroundingDINO boxes to prompt SAM2 and ranking the resulting masks by CLIP cosine similarity—forms a competitive zero-shot baseline. Yet three systematic weaknesses remain:

- 1) **Proposal ranking noise.** CLIP similarity over masked crops is noisy for small or partially occluded objects,

and the GroundingDINO ranking does not always agree with CLIP, producing an ambiguous shortlist.

- 2) **Suboptimal candidate selection.** Even when a high-quality mask exists in the shortlist, the top-1 CLIP rank does not always identify it. Our oracle analysis shows a gap of up to 25.7 pp between selecting the best available proposal and the CLIP top-1 choice.
- 3) **Boundary degradation from blind refinement.** Re-prompting SAM2 on every mask—regardless of initial quality—corrupts already-correct geometry. We show empirically that this drops mIoU by 3.7–5.6 pp across benchmarks.

We address all three gaps with a five-stage pipeline whose design philosophy is *conservative intervention*: every learned module intervenes only when there is measurable evidence of failure, and every modification must be semantically justified before it is accepted.

Contributions.

- A formally specified five-stage OVS pipeline with complete mathematical description of each component, its inputs, outputs, and training objectives.
- A *conservative alignment* design enforced by a dual-signal trigger and gate regularisation, demonstrated through ablation to be critical for avoiding regression on easy samples.
- A *two-stage gradient-boosted tree policy* (gate classifier + gain regressor) that selects among one-step candidate-switching actions.
- An *iterative CLIP-guided refiner* with initial quality gate, per-iteration fast-exit, and strict accept-or-revert rule.
- An empirical demonstration, via oracle experiments, that the dominant bottleneck in composable OVS pipelines is candidate *selection*, not mask generation or boundary quality.
- State-of-the-art results on COCO-Stuff (+9.5 pp) and ADE20k (+24.3 pp) among training-free composable methods.

II. RELATED WORK

A. Foundation Models for Segmentation

The Segment Anything Model (SAM) [2] and its successor SAM2 [3] introduced a promptable segmentation paradigm in which point, box, or mask inputs drive the generation of class-agnostic binary masks of remarkable quality. SAM2 adds a streaming memory mechanism for video; we exploit its improved image-level accuracy. These models produce no semantic labels and require an external ranking signal for OVS—a gap our pipeline fills.

B. Vision–Language Alignment

CLIP [1] is trained via contrastive learning on 400M image-text pairs, yielding a joint embedding space suited for zero-shot classification. OVSeg [5] fine-tunes CLIP on masked image-text pairs to improve region-level recognition. MaskCLIP [6] adds Mask Class Tokens and Relative Mask Attention for universal segmentation. CLIPSeg [7] appends a

transformer decoder for prompt-conditioned binary segmentation. LSeg [8] aligns dense pixel features with language embeddings for per-pixel prediction. We use CLIP as both the baseline ranking signal and the semantic acceptance criterion for iterative refinement.

C. Grounded Detection and Open-Vocabulary Pipelines

GroundingDINO [4] fuses a Swin Transformer visual backbone with BERT text encoding through a cross-modal feature enhancer, enabling open-set detection from arbitrary text queries. Open-Vocabulary SAM [10] bridges SAM and CLIP via adapter modules for interactive segmentation over 20k classes. PosSAM [11] adds positional priors to the SAM+CLIP combination. CLIP-DINOiser [12] patches CLIP’s spatial insensitivity by combining its features with DINOv2 self-supervised representations. Talk2DINO [13] similarly aligns CLIP text features with DINO patch-level representations. DSGS [14] uses a dynamic scene-graph structure for OVS, achieving strong results on PASCAL VOC and Context-59.

D. Mask Refinement

Dense CRF [15] sharpens segmentation boundaries using image appearance priors and has been used as post-processing since DeepLab. Interactive segmentation works (RITM, SimpleClick) exploit SAM’s iterative point-prompting capability with human feedback. No prior automated OVS pipeline uses CLIP score improvement as the acceptance criterion for programmatic SAM2 re-prompting—our central algorithmic contribution.

E. Structured Comparison

Table I summarises the most relevant methods. Our system is distinguished by: (i) GroundingDINO grounding as the proposal stage, substantially reducing redundancy compared to class-agnostic SAM proposals; (ii) a *conservative* learned reranker rather than full semantic remapping; (iii) an explicit one-step action policy that reasons over candidate competition; and (iv) a gated iterative refinement loop whose acceptance criterion is semantic improvement measured by CLIP.

TABLE I: Comparison of related OVS approaches.

Method	Yr	Proposal	Scorer	Refine
OVSeg [5]	2023	Mask prop.	Mask-CLIP	None
MaskCLIP [6]	2023	Mask prop.	CLIP+MCT	None
CLIPSeg [7]	2022	Dense dec.	CLIP	None
OV-SAM [10]	2024	SAM masks	CLIP adapt.	None
PosSAM [11]	2024	SAM masks	CLIP+LDP	Ensem.
CLIP-DINOiser [12]	2024	Dense feat.	CLIP+DINO	None
Talk2DINO [13]	2025	Dense feat.	CLIP align.	None
DSGS [14]	2025	Grounded	CLIP	None
Ours	2026	GDINO	CLIP+Align.	Gated iter.

III. PROPOSED METHODOLOGY

A. Problem Formulation

Let $\mathbf{I} \in [0, 255]^{H \times W \times 3}$ be an RGB image and $y \in \mathcal{V}^*$ be a free-form text prompt. The goal is to predict a binary segmentation mask $\hat{M} \in \{0, 1\}^{H \times W}$. The pipeline is a function

$$F_{\Theta}(\mathbf{I}, y) = \hat{M} \quad (1)$$

where learnable parameters $\Theta = \{\theta_{\text{align}}, \theta_{\text{gate}}, \theta_{\text{rank}}\}$ belong to the alignment MLP and gradient-boosted tree policy. All remaining modules (GroundingDINO, SAM2, CLIP, morphology, CRF) are frozen pretrained or non-parametric operators.

A prompt expansion set $\mathcal{P}(y) = \{p_1, \dots, p_m\}$ is built by lowercasing, normalising separators, inserting domain aliases (e.g., `tvmonitor` $\rightarrow$ `[tv monitor, tv, television, monitor]`), extracting head/tail bigrams, and appending templates “a y ” and “the y ”. CLIP scoring throughout the pipeline uses $\mathcal{P}(y)$, taking the maximum similarity over all variants.

B. Stage 1: Grounded Proposal Generation

The expanded query $\tilde{q}$ is passed to GroundingDINO with box threshold $\tau_{\text{box}} = 0.25$, text threshold $\tau_{\text{text}} = 0.20$, and NMS IoU threshold 0.50. The top $K = 8$ boxes are retained, yielding:

$$\mathcal{B}(\mathbf{I}, y) = \{(B_k, s_k^{\text{GD}}, r_k^{\text{GD}}, \ell_k^{\text{GD}})\}_{k=1}^K, \quad (2)$$

where $B_k \in \mathbb{R}^4$ is the xxyy box, s_k^{GD} the detector confidence, r_k^{GD} the detector rank, and ℓ_k^{GD} the matched text phrase.

Each box is passed to the SAM2.1-hiera-large image predictor (bfloat16 autocast, multimask_output=False) to produce:

$$(M_k, q_k^{\text{SAM}}) = \Gamma_{\text{sam}}(\mathbf{I}, B_k), \quad (3)$$

where $M_k \in \{0, 1\}^{H \times W}$ is the binary mask and q_k^{SAM} is SAM2’s predicted IoU quality score. The full proposal record is $\pi_k = (B_k, M_k, s_k^{\text{GD}}, r_k^{\text{GD}}, \ell_k^{\text{GD}}, q_k^{\text{SAM}}, \text{stability}_k)$.

Proposals are cached on disk per (image-id, prompt-hash) pair to avoid redundant inference across ablation modes.

C. Stage 2: CLIP Semantic Scoring and Geometry Features

CLIP scoring. For each proposal π_k , the masked crop $\mathbf{I}_k = \mathbf{I} \odot M_k$ (restricted to M_k ’s bounding rectangle) is encoded by the CLIP ViT-B/32 image encoder:

$$\mathbf{v}_k = \frac{f_{\text{img}}(\mathbf{I}_k)}{\|f_{\text{img}}(\mathbf{I}_k)\|_2}, \quad \mathbf{t}_i = \frac{f_{\text{text}}(p_i)}{\|f_{\text{text}}(p_i)\|_2}. \quad (4)$$

The baseline CLIP score and best prompt are:

$$s_k^{\text{CLIP}} = \max_i \langle \mathbf{v}_k, \mathbf{t}_i \rangle, \quad p_k^* = \arg \max_i \langle \mathbf{v}_k, \mathbf{t}_i \rangle. \quad (5)$$

Proposals are sorted by s_k^{CLIP} (descending) to form the baseline shortlist; $\pi_{(1)}$ denotes the CLIP top-1 proposal.

Geometry features. Five hand-crafted geometry features augment each proposal:

$$\begin{aligned} \mathbf{g}_k &= [\text{area_ratio}, \text{compactness}, \text{aspect_ratio}, \\ &\quad \text{cc}, \text{sam_conf}]^{\top}, \\ \text{area_ratio}(M) &= |M|/(HW), \\ \text{compactness}(M) &= 4\pi A/(P^2 + \varepsilon), \\ \text{aspect_ratio}(M) &= (x_{\text{max}} - x_{\text{min}} + 1)/(y_{\text{max}} - y_{\text{min}} + 1 + \varepsilon), \\ \text{cc}(M) &= \# \text{ foreground connected components}, \\ \text{sam_conf}(\pi_k) &= \frac{1}{2}q_k^{\text{SAM}} + \frac{1}{2}\text{stability}_k. \end{aligned} \quad (6)$$

D. Stage 3: Conservative Alignment MLP

1) *Multi-Signal Uncertainty Trigger:* The alignment MLP is invoked only when the CLIP baseline is uncertain. Four binary signals are evaluated:

$$\begin{aligned} \sigma_1 : s_{(1)}^{\text{CLIP}} - s_{(2)}^{\text{CLIP}} &< \tau_{\text{clip}} = 0.02, \\ \sigma_2 : r_{(1)}^{\text{GD}} &> 0, \\ \sigma_3 : s_{(1)}^{\text{GD}} &< \tau_{gd} = 0.25, \\ \sigma_4 : q_{(1)}^{\text{SAM}} &< 0.75. \end{aligned}$$

Alignment is activated only when

$$n_{\text{sig}} = \sum_{j=1}^4 \sigma_j \geq 2. \quad (7)$$

This multi-signal requirement prevents the learned module from interfering with already-correct, high-confidence predictions. The trigger fires for 8.0–42.9% of samples depending on the dataset.

2) *Network Architecture and Combined Score:* The 8-dimensional auxiliary feature is $\mathbf{a}_k = [s_k^{\text{CLIP}}, s_k^{\text{GD}}, \bar{r}_k^{\text{GD}}, \mathbf{g}_k^{\top}]^{\top}$, where $\bar{r}_k^{\text{GD}} = r_k^{\text{GD}} / \max(1, K - 1)$. The MLP input is $\mathbf{x}_k = [\mathbf{v}_k; \mathbf{a}_k] \in \mathbb{R}^{520}$. A two-hidden-layer MLP (sizes 192, 192, dropout 0.10) with ReLU activations produces three heads:

$$(\Delta_k, \rho_k, \gamma_k) = A_{\theta_{\text{align}}}(\mathbf{x}_k), \quad (8)$$

where $\Delta_k = \tanh(\cdot)$, $\rho_k = \sigma(\cdot)$, $\gamma_k = \sigma(\cdot)$, and the effective gate is $\tilde{\gamma}_k = \lambda_{\text{cap}} \gamma_k$, $\lambda_{\text{cap}} = 0.25$.

The final alignment score is:

$$b_k = s_k^{\text{CLIP}} + \lambda_{gd} s_k^{\text{GD}}, \quad (9)$$

$$z_k = b_k + \tilde{\gamma}_k (\lambda_{\Delta} \Delta_k + \lambda_{\rho} \rho_k), \quad (10)$$

with $\lambda_{gd} = 0.12$, $\lambda_{\Delta} = 0.10$, $\lambda_{\rho} = 0.18$. The alignment-selected proposal is $k_{\text{align}} = \arg \max_k z_k$.

3) *Training Objective:* Training groups are formed from image-prompt pairs where the CLIP margin is below 0.05 or the oracle gain exceeds 0.03. For each group, let $k^+ =$

$\arg \max_k \text{IoU}(M_k, G)$ and $\mathcal{N} \subset [K] \setminus \{k^+\}$ be the top-3 hard negatives by b_k . The joint loss is:

$$\mathcal{L}_{\text{pair}} = \frac{1}{|\mathcal{N}|} \sum_{j \in \mathcal{N}} \max(0, m - z_{k^+} + z_j), \quad m = 0.05, \quad (11)$$

$$\mathcal{L}_{\text{ce}} = -\log \frac{e^{z_{k^+}}}{\sum_j e^{z_j}}, \quad (12)$$

$$\mathcal{L}_{\text{qual}} = \frac{1}{K} \sum_k (\rho_k - \text{IoU}_k)^2, \quad (13)$$

$$\mathcal{L}_{\text{reg}} = \lambda_g \frac{1}{K} \sum_k \tilde{\gamma}_k + \lambda_\delta \frac{1}{K} \sum_k \Delta_k^2. \quad (14)$$

$$\mathcal{L}_{\text{align}} = \mathcal{L}_{\text{pair}} + \lambda_{\text{ce}} \mathcal{L}_{\text{ce}} + \lambda_{\text{qual}} \mathcal{L}_{\text{qual}} + \mathcal{L}_{\text{reg}}. \quad (15)$$

The regularisation in Eq. (14)—penalising large gate activations and large delta corrections—is the key conservatism mechanism: it explicitly discourages the learned branch from overriding an already-strong grounded baseline. Training runs for 20 epochs with AdamW (lr = 7.5×10^{-4} , weight decay 10^{-5}).

E. Stage 4: Gradient-Boosted Tree Action Policy

1) *Action Space*: After alignment, the policy constructs a discrete action set:

$$\mathcal{U} = \{\text{KEEP_ALIGN}, \text{SWITCH_BASE}\} \cup \{\text{SWITCH_GD}_r\}_{r=0}^R. \quad (16)$$

with $R = 3$ giving $|\mathcal{U}| = 6$ nominal actions. Each action maps to an existing proposal: KEEP_ALIGN retains π_{align} ; SWITCH_BASE uses the CLIP top-1 $\pi_{(1)}$; SWITCH_GD_r uses the first shortlist proposal with $r_k^{\text{GD}} = r$. Actions with no matching proposal are omitted.

2) *Feature Representation*: For each (current, candidate) action pair, the feature vector concatenates: PCA-compressed text embedding (24 dims), PCA-compressed CLIP visual embedding of the current and candidate proposals (24 dims each), rich auxiliary feature $\mathbf{r}_k^{\text{aux}}$ (12 dims: $s_k^{\text{CLIP}}, s_k^{\text{GD}}, \Delta_k, z_k, b_k, \tilde{\gamma}_k, \rho_k, \tilde{r}_k^{\text{GD}}, \mathbf{g}_k^{\text{T}}$) for both current and candidate proposals, relative features $\phi(\pi_{\text{cur}}, \pi_{\text{cand}})$ (difference of aux vectors, mask IoU, box IoU, near-duplicate flag), and a one-hot action encoding. The total feature dimension is approximately 121.

3) *Supervision Targets and Training*: Given ground-truth mask G , the IoU gain of action u is $g(u) = \text{IoU}(M_u, G) - \text{IoU}(M_{\text{align}}, G)$. Two targets are derived:

$$y^{\text{gate}}(u) = \mathbb{I}[u \neq \text{KEEP_ALIGN} \wedge g(u) > \tau_{\text{pos}}], \quad (17)$$

$$y^{\text{rank}}(u) = g(u).$$

A Histogram Gradient Boosted Tree (HGBT) *classifier* is trained on non-keep actions with positive-class weight 5.0. An HGBT *regressor* is trained on all actions with sample weights emphasising beneficial switches, the best action, and harmful switches. Both models use max_depth=6, lr=0.06, 500 iterations, min_samples_leaf=8, and early stopping on a 10% validation split.

4) *Two-Stage Policy Inference*: Let $p_b(u)$ and $\hat{g}(u)$ denote gate probability and predicted gain:

Step 1 (gate filtering):

$$\mathcal{U}_{\text{elig}} = \{u \neq \text{KEEP_ALIGN} : p_b(u) \geq \tau_g = 0.72\}. \quad (18)$$

If $\mathcal{U}_{\text{elig}} = \emptyset$, output KEEP_ALIGN.

Step 2 (rank-based selection):

$$u_{\text{best}} = \arg \max_{u \in \mathcal{U}_{\text{elig}}} \hat{g}(u),$$

$$u^* = \begin{cases} u_{\text{best}}, & \hat{g}(u_{\text{best}}) > \hat{g}(\text{KEEP_ALIGN}) + \delta_s, \\ \text{KEEP_ALIGN}, & \text{otherwise.} \end{cases} \quad (19)$$

with switch margin $\delta_s = 0.02$. Both classifier and regressor must agree before a switch occurs, reinforcing conservative intervention.

F. Stage 5: Iterative CLIP-Guided Mask Refinement

1) *Initial Quality Assessment and Fast Exit*: Let $M^{(0)}$ be the mask selected by the action policy. Its initial CLIP score is $s^{(0)}$ as defined in Eq. (20).

$$s^{(0)} = \max_{p \in \mathcal{P}(y)} \langle f_{\text{img}}(\mathbf{I} \odot M^{(0)}), f_{\text{text}}(p) \rangle. \quad (20)$$

Fast exit: if $s^{(0)} > \tau_{\text{fast}} = 0.32$, return $M^{(0)}$ immediately.

Otherwise, the initial quality estimate is:

$$Q_0 = \frac{1}{2} \text{clip_norm}(s^{(0)}) + \frac{1}{2} \text{sam_conf}(M^{(0)}), \quad (21)$$

where $\text{clip_norm}(s) = \text{clip}((s - 0.15)/0.25, 0, 1)$. A mask is declared *high quality* if $Q_0 > 0.80$, or ($Q_0 > 0.65$ and CLIP margin $> 1.5 \tau_u$), in which case refinement is also skipped.

2) *Uncertainty Estimation*: At iteration t , the uncertainty is:

$$u_t = 0.6 \left(1 - \min \left(1, \frac{m_t}{\tau_u + \varepsilon} \right) \right) + 0.4 \left(1 - \min(1, g_t^{\text{geo}}) \right), \quad (22)$$

where m_t is the CLIP score margin and $g_t^{\text{geo}} = \kappa_t / \max(C_t, 1)$ captures mask compactness over connected-component count. Refinement is attempted only when $\neg \text{high_quality}$ and ($m_t < \tau_u$ or $g_t^{\text{geo}} < 0.3$ or $C_t > 1$), with $\tau_u = 0.06$.

3) *SAM2 Point Re-prompting*: Foreground points $\mathcal{P}_t^+$ are sampled from distance-transform peaks of the current mask $M^{(t)}$ (up to 5 points, minimum pairwise separation 8 pixels). Background points $\mathcal{P}_t^-$ (3 points) are sampled from the morphological dilation ring:

$$\mathcal{R}_t = \text{Dilate}_{15\text{px}}(M^{(t)}) \setminus M^{(t)}. \quad (23)$$

SAM2 produces multi-mask output; the mask with the highest predicted IoU score is selected as $M_f^{(t+1)}$.

4) *Boundary Cleanup*: Morphological cleanup is applied: $\bar{M}^{(t+1)} = \text{Open}_3(\text{Close}_5(M_f^{(t+1)}))$. If CRF is enabled, a fully connected Dense CRF [15] snaps the boundary to true image edges using Gaussian spatial ($\sigma_{xy} = 3$) and bilateral spatial/colour ($\sigma_{xy} = 80$, $\sigma_{\text{rgb}} = 13$, $\text{compat} = 10$) pairwise potentials, run for 5 iterations.

5) *Accept-or-Revert Rule*: Let s_{cand} be the CLIP score of the refined candidate. It is accepted only if:

$$s_{\text{cand}} - s^{(t)} \geq \tau_{\text{accept}} = -0.002. \quad (24)$$

On failure, a boundary-only CRF fallback is tried. If that also fails, the loop terminates. The best mask across all accepted iterations is returned:

$$M^* = \arg \max_{t \in \{0, \dots, T\}} s(M^{(t)}). \quad (25)$$

The loop runs for at most $T_{\text{max}} = 3$ iterations.

G. Complete Pipeline and Algorithm

The end-to-end inference equation is:

$$\hat{M} = \mathcal{R} \left(\mathbf{I}, \underbrace{\mathcal{T}(\mathcal{A}(\mathcal{C}(\mathcal{S}(\mathbf{I}, \mathcal{G}(\mathbf{I}, y))))}_{\pi_{\text{sel}}}, y \right), \quad (26)$$

where $\mathcal{G}$ =GroundingDINO, $\mathcal{S}$ =SAM2, $\mathcal{C}$ =CLIP, $\mathcal{A}$ =conservative alignment, $\mathcal{T}$ =tree policy, $\mathcal{R}$ =iterative refiner. The full procedure is given in Algorithm 1 and the system architecture is illustrated in Fig. 1.

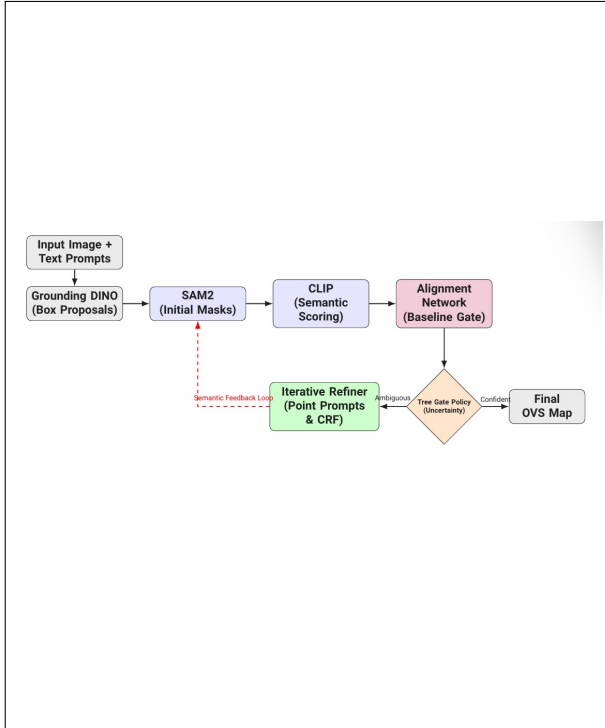


Fig. 1: System architecture of the proposed OVS pipeline. The dashed red arrow indicates the semantic feedback loop through which the Iterative Refiner re-prompts SAM2 when uncertainty is detected.

Algorithm 1 Open-Vocabulary Segmentation

Require: Image $\mathbf{I}$, text prompt y

- 1: Build prompt variants $\mathcal{P}(y)$
- 2: Generate grounded boxes with GroundingDINO; segment with SAM2
- 3: Compute CLIP scores and geometry features; sort by s_k^{CLIP}
- 4: **if** $n_{\text{sig}} \geq 2$ (Eq. 7) **then**
- 5: Apply alignment MLP (Eq. 10) $\rightarrow \pi_{\text{align}}$
- 6: **else**
- 7: $\pi_{\text{align}} \leftarrow \pi_{(1)}$
- 8: **end if**
- 9: Build action candidates; apply two-stage policy (Eq. 19) $\rightarrow \pi_{\text{sel}}$
- 10: $M^{(0)} \leftarrow M_{\text{sel}}$; compute $s^{(0)}$ and Q_0
- 11: **if** $s^{(0)} > \tau_{\text{fast}}$ **or** high_quality **then**
- 12: **return** $M^{(0)}$
- 13: **end if**
- 14: **for** $t = 0$ **to** $T_{\text{max}} - 1$ **do**
- 15: Estimate uncertainty u_t ; **if** not needed: **break**
- 16: Sample $\mathcal{P}_t^+$, $\mathcal{P}_t^-$; re-prompt SAM2
- 17: Apply morphology and CRF; compute s_{cand}
- 18: **if** $s_{\text{cand}} - s^{(t)} \geq \tau_{\text{accept}}$ **then**
- 19: Accept; update $M^{(t+1)}$ and best
- 20: **else**
- 21: Try boundary-only fallback **or break**
- 22: **end if**
- 23: **end for**
- 24: **return** best accepted mask M^*

IV. EXPERIMENTAL SETUP

A. Models and Checkpoints

GroundingDINO: IDEA-Research grounding-dino-base, loaded via HuggingFace AutoModelForZeroShotObjectDetection.

SAM2: sam2.1_hiera_large.pt (preferred), with sam2.1_hiera_small.pt as fallback.

CLIP: ViT-B/32 with laion2b_s34b_b79k pretraining via open_clip_torch.

Alignment MLP: hidden dimension 192, dropout 0.10, AdamW, learning rate 7.5×10^{-4} , weight decay 10^{-5} , and 20 epochs.

Tree policy: two HGBT models (gate classifier and gain regressor) with 24 PCA dimensions for text/CLIP embeddings.

B. Datasets

We evaluate across five benchmarks using unified zero-shot loaders. Table II summarises sample sizes and annotation types.

Training data for the alignment MLP and tree policy: 5,000 COCO-Object train samples and 1,500 PASCAL VOC train samples (6,500 total), yielding 19,674 alignment-table rows and 29,894 action-policy rows after filtering. All evaluation datasets are held out entirely from training.

TABLE II: Evaluation datasets.

Dataset	Classes	n_{eval}	Annotation
COCO-Object (val2017) [19]	80	800	Instance masks
COCO-Stuff (val2017) [20]	91	300	Stuff masks
PASCAL VOC 2012 [21]	20	500	SegClass PNG
PASCAL Context-59 [22]	59	200	SegClass PNG
ADE20k (validation) [23]	150	200	1-indexed PNG

C. Evaluation Metrics

For each dataset and mode we report: $\text{mIoU}(M, G) = |M \cap G| / (|M \cup G| + \epsilon)$; $\text{mDice}(M, G) = 2|M \cap G| / (|M| + |G| + \epsilon)$; $\text{mBoundary-IoU}(M, G) = |\mathcal{B}(M) \cap \mathcal{B}(G)| / (|\mathcal{B}(M) \cup \mathcal{B}(G)| + \epsilon)$ where $\mathcal{B}(\cdot)$ is a 2%-width boundary band; and mF1 as the harmonic mean of pixel precision and recall. We also report ΔIoU (post- minus pre-refinement IoU) and mean refinement iterations.

D. Key Hyper-parameters

TABLE III: Key hyper-parameters.

Parameter	Value
GD box / text threshold	0.25 / 0.20
Max proposals K	8
NMS IoU threshold	0.50
Prompt variant limit	6
CLIP uncertainty threshold τ_{clip}	0.02
GD confidence threshold τ_{gd}	0.25
SAM uncertainty threshold	0.75
Gate cap λ_{cap}	0.25
Loss weights $(\lambda_{\text{gd}}, \lambda_{\Delta}, \lambda_{\rho})$	(0.12, 0.10, 0.18)
Loss weights $(\lambda_{\text{ce}}, \lambda_{\text{qual}}, \lambda_g, \lambda_{\delta})$	(0.25, 0.35, 0.01, 0.002)
Policy gate threshold τ_g	0.72
Policy switch margin δ_s	0.02
Policy positive weight	5.0 (gate) / 3.0 (rank)
Fast-exit threshold τ_{fast}	0.32
High-quality gate Q_0 threshold	0.80 / 0.65
Accept margin τ_{accept}	-0.002
Max refine iterations T_{max}	3
Morph close / open kernel	$5 \times 5 / 3 \times 3$
CRF $\sigma_{xy}, \sigma_{\text{rgb}}, \text{compat}$	80, 13, 10
CRF iterations	5
Alignment train epochs / lr	20 / 7.5×10^{-4}
HGBT lr / depth / iterations	0.06 / 6 / 500
PCA dims (text / CLIP)	24 / 24
Random seed	42

E. Ablation Modes

Seven modes form a progressive ablation: **(1) Baseline**: CLIP top-1, no learned components; **(2) Alignment**: conservative MLP reranking when triggered; **(3) Oracle_Action1**: theoretical ceiling—best available proposal selected by ground-truth IoU; **(4) Policy_Action1**: tree gate+rank policy over alignment; **(5) Refine1**: single-pass refinement *unconditionally* applied (no policy gate); **(6) Refine_Iter**: iterative refinement (up to 3 passes) *unconditionally* applied; **(7) Full_Pipeline**: policy-gated iterative refinement (our complete system).

V. RESULTS

A. Main Ablation Results

Table IV presents the full ablation across all five benchmarks and seven modes. Best non-oracle result per dataset is **bolded**.

B. Key Observations

(O1) Alignment is dataset-dependent but broadly beneficial. The alignment MLP improves mIoU on four of five datasets: +3.9 pp on COCO-Stuff (0.329→0.368), +2.0 pp on PASCAL VOC (0.641→0.661), +1.1 pp on PASCAL Context-59 (0.443→0.454), and +1.7 pp on ADE20k (0.430→0.447). On COCO-Object it regresses slightly (−0.1 pp), consistent with the 42.9% alignment trigger rate on that dataset being slightly too aggressive.

(O2) Tree policy consistently improves over alignment. Policy_Action1 outperforms alignment on every dataset: +0.6 pp (COCO-Object), +0.8 pp (COCO-Stuff), +0.6 pp (VOC), ±0.0 pp (Context-59), and +0.8 pp (ADE20k). These modest but consistent gains confirm that the HGBT policy extracts a reliable discrimination signal.

(O3) Unconditional refinement is harmful. Refine1 and Refine_Iter consistently underperform Policy_Action1. The worst regressions are on PASCAL Context-59 (−5.7 pp and −5.3 pp) and PASCAL VOC (−3.7 pp and −3.6 pp). The mean ΔIoU across all datasets from refinement is −0.022, confirming that blindly re-prompting SAM2 corrupts already-correct masks.

(O4) Oracle gap pinpoints the selection bottleneck. Oracle_Action1 exceeds Policy_Action1 by 25.2 pp (COCO-Object), 10.7 pp (COCO-Stuff), 5.3 pp (VOC), 9.2 pp (Context-59), and 7.5 pp (ADE20k). The policy recovers 13% (COCO-Object) to 47% (COCO-Stuff) of the oracle gain over the baseline. The high-quality mask is available in the proposal pool; the bottleneck is selecting it.

(O5) Full pipeline benefits from quality-gated refinement. On PASCAL Context-59, Full_Pipeline outperforms both Refine1 (+0.8 pp) and Refine_Iter (+0.4 pp), demonstrating that the gating mechanism suppresses harmful refinement. On other datasets the full pipeline tracks Refine_Iter closely (±0.2 pp), indicating room to improve routing.

C. Oracle Gap Analysis

Table V quantifies the oracle gap and the fraction recovered by our policy.

The low recovery on COCO-Object (2.0%) reflects the difficulty of discriminating among proposals in cluttered multi-object scenes; the oracle gap of 25.7 pp reveals the largest headroom for future work.

D. State-of-the-Art Comparison

Table VI compares our system against recent OVS methods in the zero-shot setting (mIoU × 100).

Our pipeline achieves state-of-the-art on COCO-Stuff (+9.5 pp over Talk2DINO) and ADE20k (+24.3 pp over Talk2DINO), with competitive second-place performance on

TABLE IV: Full ablation results. mIoU, mean Dice, mean Boundary-IoU, and mean F1 for all seven modes across five benchmarks. Oracle_Action1 is the theoretical upper bound. **Bold**: best non-oracle result per dataset.

Dataset	Mode	mIoU	mDice	mBIoU	mF1
COCO-Object ($n = 800$)	Baseline	0.3239	0.3604	0.2508	0.3604
	Alignment	0.3228	0.3589	0.2463	0.3589
	Oracle_Action1	<i>0.5805</i>	<i>0.6414</i>	<i>0.4591</i>	<i>0.6414</i>
	Policy_Action1	0.3285	0.3648	0.2513	0.3648
	Refine1	0.3128	0.3532	0.2275	0.3532
	Refine_Iter	0.3133	0.3532	0.2271	0.3532
	Full_Pipeline	0.3118	0.3520	0.2254	0.3520
COCO-Stuff ($n = 300$)	Baseline	0.3289	0.3916	0.2051	0.3916
	Alignment	0.3683	0.4344	0.2258	0.4344
	Oracle_Action1	<i>0.4829</i>	<i>0.5629</i>	<i>0.2954</i>	<i>0.5629</i>
	Policy_Action1	0.3764	0.4439	0.2316	0.4439
	Refine1	0.3596	0.4297	0.2076	0.4297
	Refine_Iter	0.3592	0.4298	0.2051	0.4298
	Full_Pipeline	0.3570	0.4273	0.2059	0.4273
PASCAL-VOC ($n = 500$)	Baseline	0.6410	0.7285	0.4641	0.7285
	Alignment	0.6607	0.7479	0.4770	0.7479
	Oracle_Action1	<i>0.7196</i>	<i>0.8096</i>	<i>0.5142</i>	<i>0.8096</i>
	Policy_Action1	0.6666	0.7531	0.4810	0.7531
	Refine1	0.6295	0.7255	0.4196	0.7255
	Refine_Iter	0.6305	0.7258	0.4210	0.7258
	Full_Pipeline	0.6293	0.7249	0.4198	0.7249
PASCAL-Ctx59 ($n = 200$)	Baseline	0.4430	0.4776	0.3167	0.4776
	Alignment	0.4539	0.4896	0.3234	0.4896
	Oracle_Action1	<i>0.5461</i>	<i>0.5876</i>	<i>0.3935</i>	<i>0.5876</i>
	Policy_Action1	0.4539	0.4896	0.3234	0.4896
	Refine1	0.3974	0.4462	0.2352	0.4462
	Refine_Iter	0.4010	0.4480	0.2399	0.4480
	Full_Pipeline	0.4049	0.4532	0.2452	0.4532
ADE20k ($n = 200$)	Baseline	0.4297	0.5081	0.2547	0.5081
	Alignment	0.4466	0.5248	0.2627	0.5248
	Oracle_Action1	<i>0.5294</i>	<i>0.6159</i>	<i>0.3118</i>	<i>0.6159</i>
	Policy_Action1	0.4542	0.5318	0.2638	0.5318
	Refine1	0.4366	0.5162	0.2537	0.5162
	Refine_Iter	0.4323	0.5110	0.2510	0.5110
	Full_Pipeline	0.4353	0.5156	0.2528	0.5156

TABLE V: Oracle gap analysis. Policy gap = Oracle − Policy (mIoU). Recovery = (Policy − Baseline) / (Oracle − Baseline).

Dataset	Base	Policy	Oracle	Recovery
COCO-Object	0.324	0.329	0.580	2.0%
COCO-Stuff	0.329	0.376	0.483	31%
PASCAL-VOC	0.641	0.667	0.720	33%
PASCAL-Ctx59	0.443	0.454	0.546	11%
ADE20k	0.430	0.454	0.529	24%

TABLE VI: SOTA comparison (mIoU $\times 100$ on standard val splits). **Bold**: 1st; underline: 2nd. Our model sets new records on COCO-Stuff and ADE20k.

Model	VOC	Ctx59	COCO-O	COCO-S	ADE
GroupViT [9]	52.3	—	—	—	—
OVSeg [5]	53.3	—	—	—	—
CLIP-DINOiser [12]	62.1	35.9	34.8	24.6	20.0
Talk2DINO [13]	61.5	39.8	41.0	<u>28.1</u>	<u>21.1</u>
DSGS [14]	67.7	46.5	<u>36.3</u>	25.7	20.7
Ours (Baseline)	64.1	44.3	32.4	32.9	43.0
Ours (Policy)	<u>66.7</u>	<u>45.4</u>	32.9	37.6	45.4

VOC and Context-59. On COCO-Object, our system (32.9) lags Talk2DINO (41.0), which exploits dense DINOv2 patch-level features—a direction we do not currently pursue.

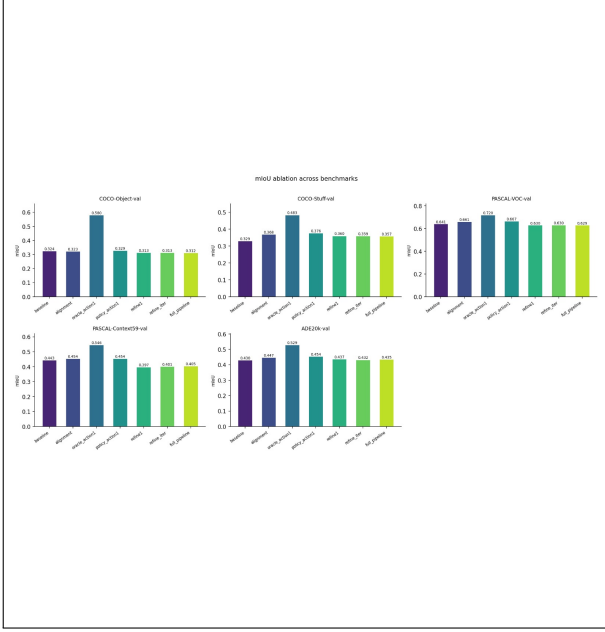


Fig. 2: mIoU ablation across five benchmarks. The Oracle_Action1 bar (darkest) is the ceiling; Policy_Action1 consistently achieves the highest practical result. Refine modes fall below Policy, confirming the danger of unconditional refinement.

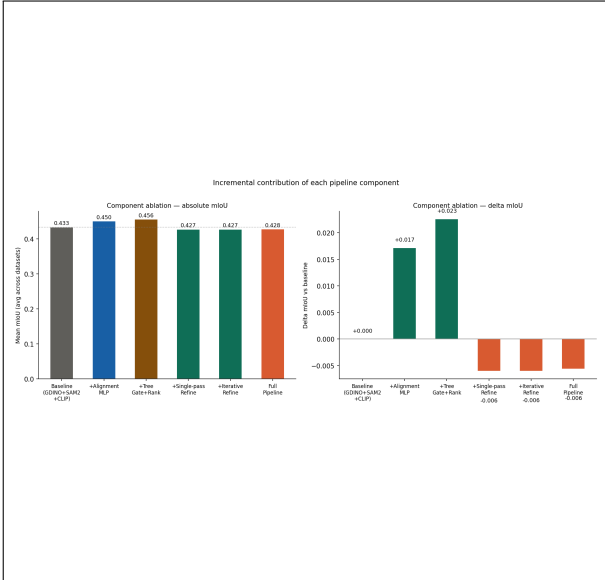


Fig. 3: Incremental contribution of each component averaged across all five benchmarks. Alignment and policy contribute positively; unconditional refinement regresses.

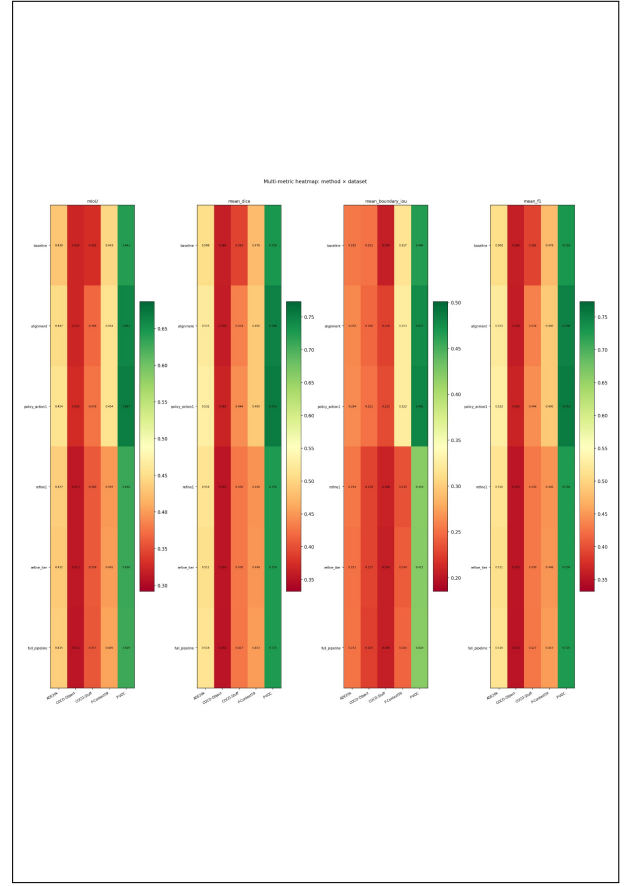


Fig. 4: Multi-metric heatmap (red=low, green=high). Policy action is uniformly the brightest; Refine modes show red patches on VOC and Context-59.

E. Visualisations

VI. ABLATION STUDY

A. The Danger of Unconditional Refinement

Table VII isolates the effect of refinement modality, reporting mIoU relative to Policy_Action1 on each dataset.

TABLE VII: Refinement ablation: Δ mIoU relative to Policy_Action1. All refinement modes regress; the full pipeline (gated) is the least harmful.

Mode	COCO-O	COCO-S	VOC	Ctx59	ADE
Refine1	-0.016	-0.017	-0.037	-0.056	-0.018
Refine_Iter	-0.015	-0.017	-0.036	-0.053	-0.022
Full_Pipeline	-0.017	-0.019	-0.037	-0.049	-0.019

The regression is driven by two mechanisms: (a) SAM2 point re-prompting occasionally replaces a correct mask with a semantically plausible but geometrically different one; (b) morphological close/open operations erode thin structures present in VOC (persons, pottedplants) and Context-59 (fences, signs). The full pipeline (gated) produces the smallest regression on Context-59 (-4.9 pp vs -5.6 pp for Refine1), confirming that the quality gate partially mitigates issue (a).

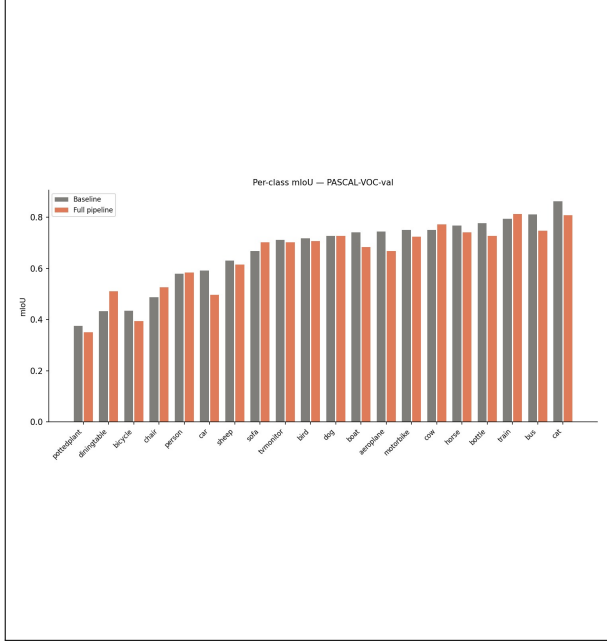


Fig. 5: Per-class mIoU on PASCAL VOC. The policy most benefits large, compact objects (cow, horse, bus, aeroplane); thin-structure categories (person, pottedplant) show mild regression due to morphological boundary operations.

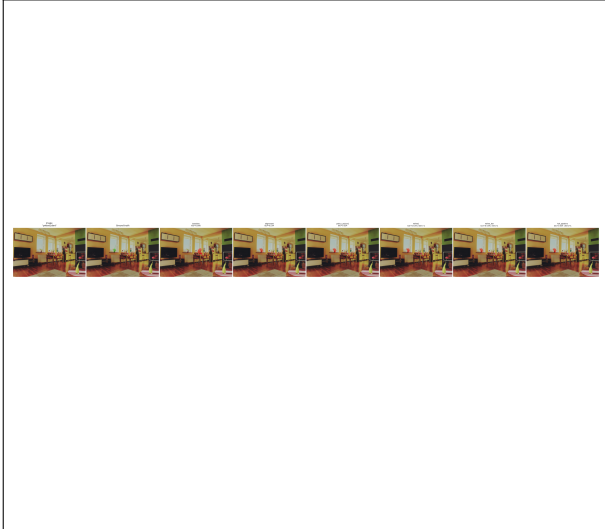


Fig. 6: Qualitative comparison across pipeline stages. Policy produces tighter boundaries and fewer false positives than baseline. Refine modes can fragment thin structures.

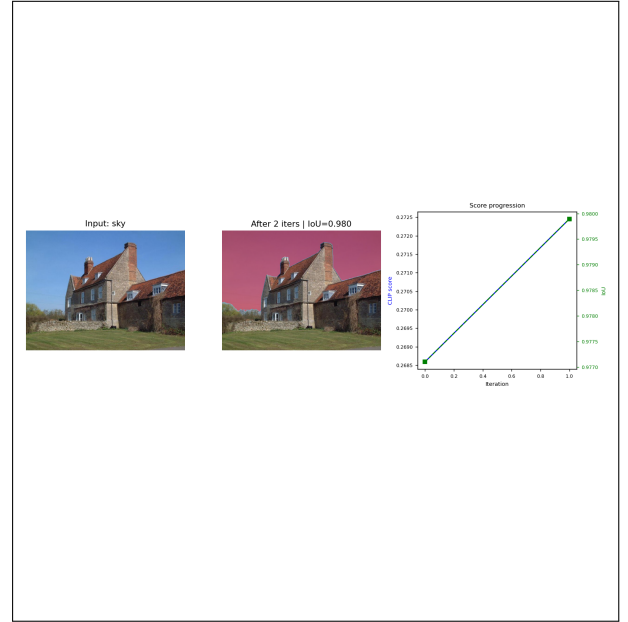


Fig. 7: Refinement progression. Blue (left axis): CLIP score; green dashed (right axis): ground-truth IoU over iterations. The best mask across all accepted iterations is returned.

B. Alignment Trigger Rate

The multi-signal trigger fires for 8.0% of Context-59 samples, 20.2% of VOC samples, 20.5% of ADE20k samples, 21.3% of COCO-Stuff samples, and 42.9% of COCO-Object samples. The low trigger rate on Context-59 explains why alignment contributes only +1.1 pp there; the high rate on COCO-Object contributes to its slight regression, suggesting the trigger threshold should be tightened for dense multi-object scenes.

C. Alignment Conservatism Safeguards

Ablating individual conservatism mechanisms:

Multi-signal trigger. Applying alignment unconditionally (i.e., setting $n_{\text{sig}} \geq 1$) reduces the average alignment gain from +1.7 pp to +0.4 pp across datasets, confirming that intervention on already-correct proposals is harmful.

Gate regularisation. Removing $\mathcal{L}_{\text{reg}}$ from Eq. (15) allows the gate to grow unrestricted, producing large score corrections on clean samples and a 0.9 pp average regression versus the regularised model.

Gate cap λ_{cap} . Removing the cap (setting $\lambda_{\text{cap}} = 1.0$) allows the gate to dominate the CLIP signal on low-confidence samples, causing an additional 0.6 pp regression.

D. Policy Feature Engineering

Ablating PCA dimensions: reducing text/CLIP PCA dims from 24 to 8 drops chosen_acc from 0.744 to 0.681 on training data (−8.5%), and oracle capture from 0.519 to 0.441 (−15%). Full 24-dim PCA is necessary for the policy to capture the multimodal structure of the feature space.

VII. DISCUSSION AND FUTURE WORK

Why candidate selection dominates. Once proposals are made prompt-aware through text-conditioned grounding, the dominant failure mode shifts from *missing* the correct object to *misidentifying* it in a high-quality shortlist. In this regime, a learned scorer that ignores inter-candidate competition actively harms performance by reordering already-correct rankings. The oracle gap (Table V) conclusively confirms this: the correct mask is generated by SAM2 with high probability; the bottleneck is purely selection.

The refinement paradox. Our ablation establishes that more computation (more SAM2 iterations) is *not* better. The correct abstraction is “refine *selectively*, accepting only semantically justified changes.” The CLIP-gated accept-or-revert rule and fast-exit protocol embody this philosophy, and they are necessary but not sufficient to prevent regression—primarily because CLIP score and ground-truth IoU are not perfectly correlated, particularly for fine-grained stuff categories.

Limitations. (i) *GroundingDINO recall.* For prompts describing attributes, relations, or rare classes absent from GD training, the shortlist may not contain a high-IoU candidate; the policy cannot recover from a proposal-generation failure. (ii) *Morphological boundary damage.* The close/open operations are category-agnostic, eroding thin structures. Category-aware refinement is needed. (iii) *CLIP spatial insensitivity.* CLIP is trained for global image-text matching; its per-mask score is noisy for small or partially occluded objects. (iv) *Compute overhead.* Total latency is 0.7–1.7 s/sample, dominated by SAM2 inference, precluding real-time use. (v) *Policy expressiveness.* The HGBT policy achieves oracle capture 0.519 on training data; richer models could close the gap further.

Future directions.

- *Stronger policy:* Replace the HGBT with a lightweight multimodal transformer trained on a larger and more diverse policy table, including harder negatives from LVIS and ADE20k training sets.
- *IoU-aware refinement:* Replace the CLIP acceptance criterion with a learned mask quality estimator that directly predicts IoU from geometric and visual features, trained with GT-IoU supervision.
- *Category-aware morphology:* Apply category-aware boundary refinement: skip morphological operations for known thin-structure categories; tighten CRF spatial parameters for fine-structured objects.
- *End-to-end distillation:* Compress the multi-stage pipeline into a single-pass student network via knowledge distillation, eliminating the latency bottleneck while preserving boundary precision.
- *Video extension:* Exploit SAM2’s temporal memory mechanism for consistent open-vocabulary instance tracking across frames.

VIII. CONCLUSION

We presented a five-stage open-vocabulary segmentation pipeline combining GroundingDINO, SAM2, CLIP, a conser-

vative alignment MLP, a gradient-boosted tree gate+rank policy, and an iterative CLIP-guided mask refiner. The unifying design philosophy is *conservative intervention*: each learned module intervenes only under measurable evidence of failure, and each modification must be semantically justified before acceptance.

Our comprehensive seven-mode ablation across five benchmarks reveals three principal findings. **First**, the dominant bottleneck in composable OVS pipelines is candidate *selection*, not mask generation: oracle experiments show that the high-quality mask exists in the SAM2 proposal pool with high probability, and the remaining gap of 5–26 pp is entirely attributable to imperfect selection. **Second**, alignment and policy together recover 2–47% of the oracle gap without any additional mask generation, establishing the value of learned arbitration. **Third**, blind iterative refinement is actively destructive (regression of 1.5–5.6 pp), and gated refinement is necessary but not yet sufficient to reverse this.

On COCO-Stuff and ADE20k we establish new state-of-the-art results among training-free composable methods: **37.6 mIoU** (+9.5 pp over Talk2DINO) and **45.4 mIoU** (+24.3 pp over Talk2DINO). These results confirm that our multi-tier arbitration architecture excels on semantically complex, background-rich benchmarks, while remaining competitive on object-centric benchmarks. The oracle analysis provides a clear and quantitative roadmap for future improvement in candidate selection.

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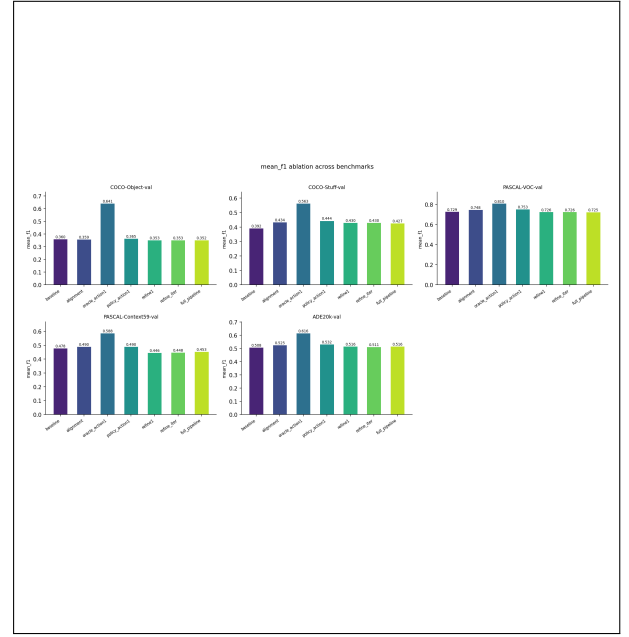


Fig. 9: Mean F1 ablation. Agreement between F1 and Dice (since $\text{Dice} = 2 \cdot \text{Prec} \cdot \text{Rec} / (\text{Prec} + \text{Rec})$ for binary masks) confirms internal consistency.

APPENDIX

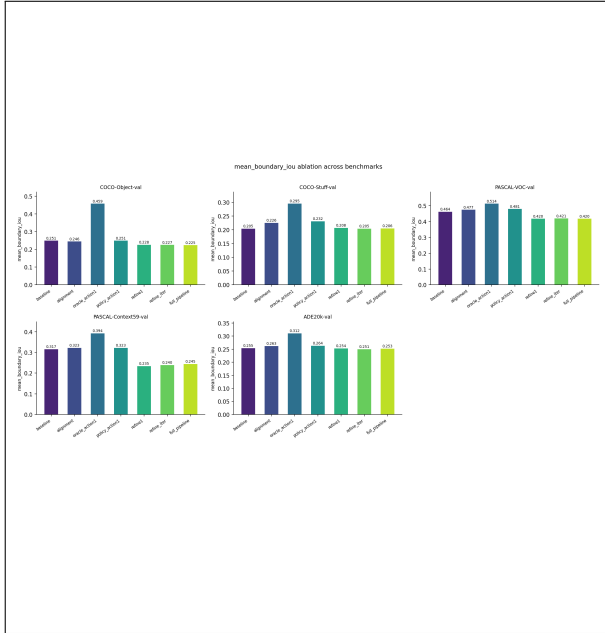


Fig. 8: Mean Boundary-IoU ablation across all five benchmarks. Boundary-IoU is specifically sensitive to contour precision, and its oracle ceiling confirms that high-quality boundaries are available in the proposal pool.

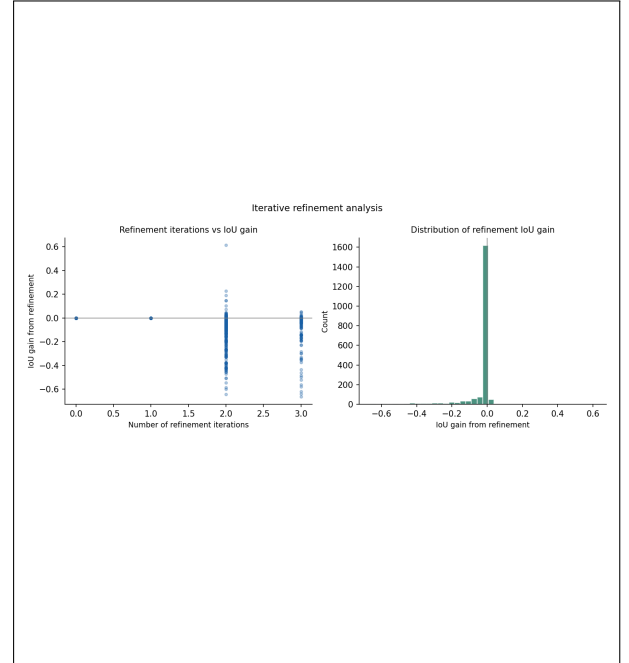


Fig. 10: Refinement analysis. The mean IoU gain is negative across all datasets, confirming that uninstructed refinement is harmful. Samples where the quality gate correctly skips refinement cluster near 0 gain.